EXPERIMENT NO. : 8 (ASSIGNMENT)

DATE: 08/10/2026

1. AIM

For the given combinational logic circuit:

1] To determine the optimal gate sizes x and y using the method of Logical Effort ($x = 10,\ y = 15$).

2] To design and simulate the sized logic circuit in 180nm CMOS technology, calculate propagation delay from transient simulation, determine the theoretical delay using logical effort, and compare them.

2. TOOL

Cadence Virtuoso

3. THEORY

1. Path Logical Effort and Branching: In multistage combinational networks, the total path effort F characterizes the total workload and is defined as the product of path logical effort $G = \prod g_i$, path branching effort $B = \prod b_i$, and path electrical effort $H = C_{out}/C_{in}$. Logical effort g accounts for the topological complexity of each gate relative to a standard inverter, while branching effort b accounts for capacitive loading on off-path fan-out branches.

2. Optimal Stage Effort and Transistor Sizing: Minimum path delay is achieved when each of the N stages bears identical stage effort $\hat{f} = F^{1/N}$. Using this optimal effort, intermediate gate sizes are uniquely determined from input to output using the stage effort relation $\hat{f} = g_i \cdot h_i = g_i \cdot \frac{C_{out,i}}{C_{in,i}}$. Transistor channel widths for each gate are then scaled proportionally to match the calculated input capacitance.

3. Path Delay Formulation: The minimum normalized path delay consists of the sum of stage effort delays and parasitic delays: $D = N \cdot \hat{f} + P$, where $P = \sum p_i$ is the total path parasitic delay. The absolute theoretical propagation delay is evaluated by multiplying the normalized delay by the unit inverter delay, $T_{p,\text{theor}} = D \cdot t_{inv}$.

4. TABLES

Table 4.1: Design Parameters

given parameter	value	unit
technology	180	nm
V_{dd}	1.8	V
channel length (L)	180	nm
input capacitance (C_{in})	8	units
load capacitance (C_L)	45	fF
reference inverter delay (t_{inv})	14.6230	ps
optimal stage effort ($\hat{f}$)	5	-

Table 4.2: Logical Effort Stage Sizing & Parameters

stage	gate type	logical effort (g)	branching effort (b)	parasitic delay (p)	size	W_n (μm)	W_p (μm)
stage 1	NAND2	1.3333	3	2	8	1.6	1.6
stage 2	NAND3	1.6667	2	3	10	2.4	1.6
stage 3	NOR2	1.6667	1	2	15	1.2	4.8

Table 4.3: Comparison of Theoretical and Practical Propagation Delay

parameter	theoretical value	practical value	unit	signed error
t_{pLH}	-	311.8570	ps	-
t_{pHL}	-	373.4920	ps	-
propagation delay (t_p)	321.7060	342.6745	ps	+6.5179%

5. CALCULATION

$$G = g_1 * g_2 * g_3$$

$$= 1.333333 * 1.666667 * 1.666667$$

$$= 3.7037043 \approx 3.7037$$

$$B = b_1 * b_2 * b_3 = 3 * 2 * 1 = 6$$

$$H = C_L / C_{in} = 45/8 = 5.625$$

$$F = G * B * H = 3.7037043 * 6 * 5.625 = 125.0000201 \approx 125$$

$$f_{hat} = 5$$

$$x = (f_{hat} * C_{in})/(g_1 * b_1)$$

$= (5 * 8)/(1.333333 * 3)$

$= 10.0000025 \approx 10$

$y = (f_{hat} * x)/(g_2 * b_2)$

$= (5 * 10)/(1.666667 * 2)$

$= 14.999997 \approx 15$

$P = p_1 + p_2 + p_3 = 2 + 3 + 2 = 7$

$D = 3 * f_{hat} + P = 3 * 5 + 7 = 22$

$T_{theor} = D * t_{inv} = 22 * 14.623 = 321.706ps$

$T_{prac} = (t_{pLH} + t_{pHL})/2 = (311.857 + 373.492)/2 = 342.6745ps$

$Error = (T_{prac} - T_{theor})/T_{theor} * 100$

$= (342.6745 - 321.706)/321.706 * 100$

$= 6.5179077\% \approx 6.52\%$

6. CIRCUIT DIAGRAMS & WAVEFORMS

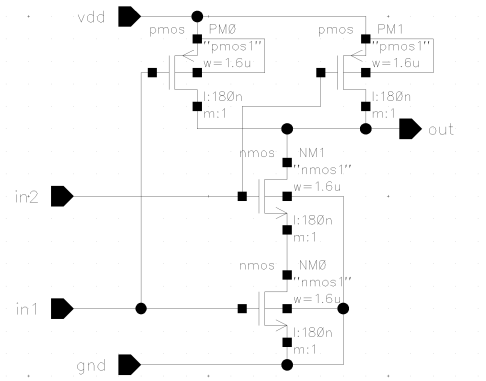


Figure 6.1: 2-input NAND (NAND2) schematic ($W_p = 1.6\mu\text{m}, W_n = 1.6\mu\text{m}$)

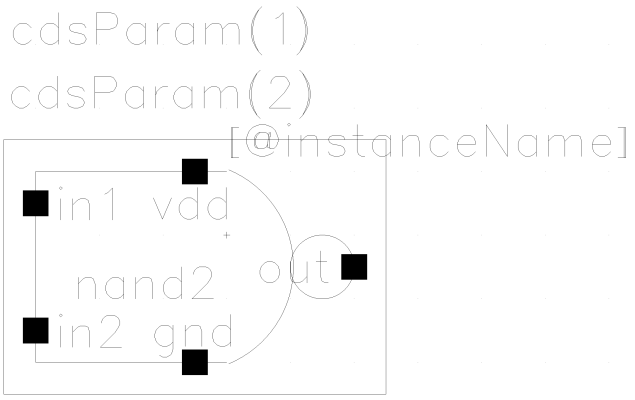


Figure 6.2: 2-input NAND (NAND2) symbol

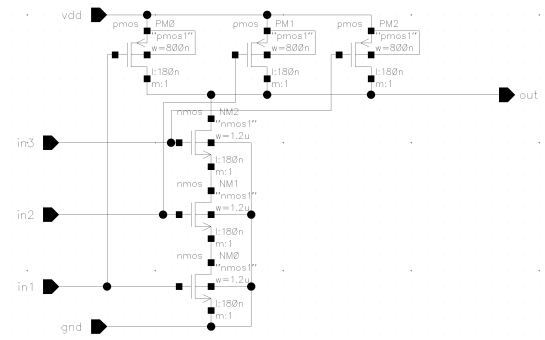


Figure 6.3: 3-input NAND (NAND3) schematic ($W_p = 800\text{nm}, W_n = 1.2\mu\text{m}$)

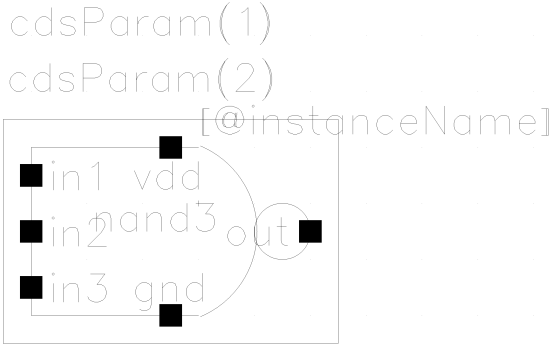


Figure 6.4: 3-input NAND (NAND3) symbol

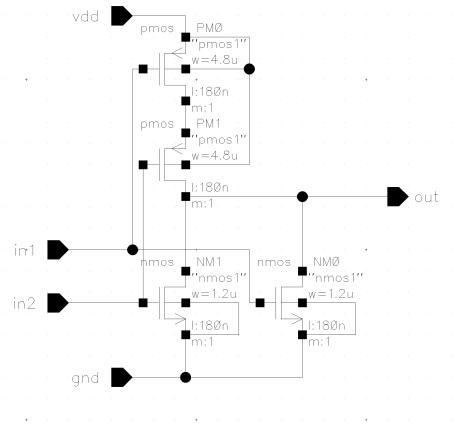


Figure 6.5: 2-input NOR (NOR2) schematic ($W_p = 4.8\mu\text{m}, W_n = 1.2\mu\text{m}$)

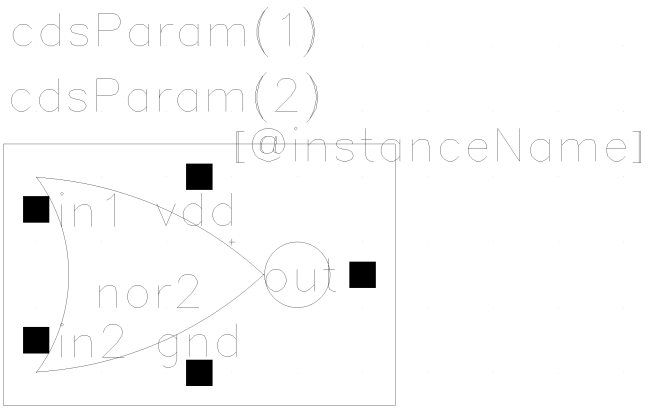


Figure 6.6: 2-input NOR (NOR2) symbol

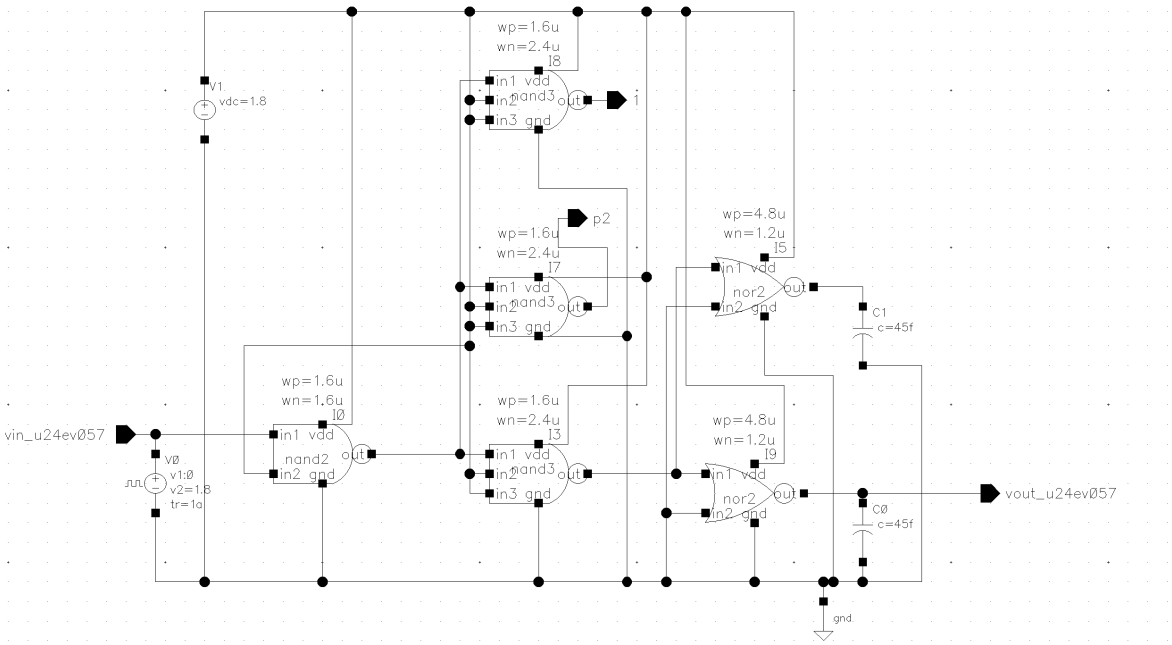


Figure 6.7: Logic block testbench circuit ($x = 10, y = 15, C_L = 45\text{fF}$)

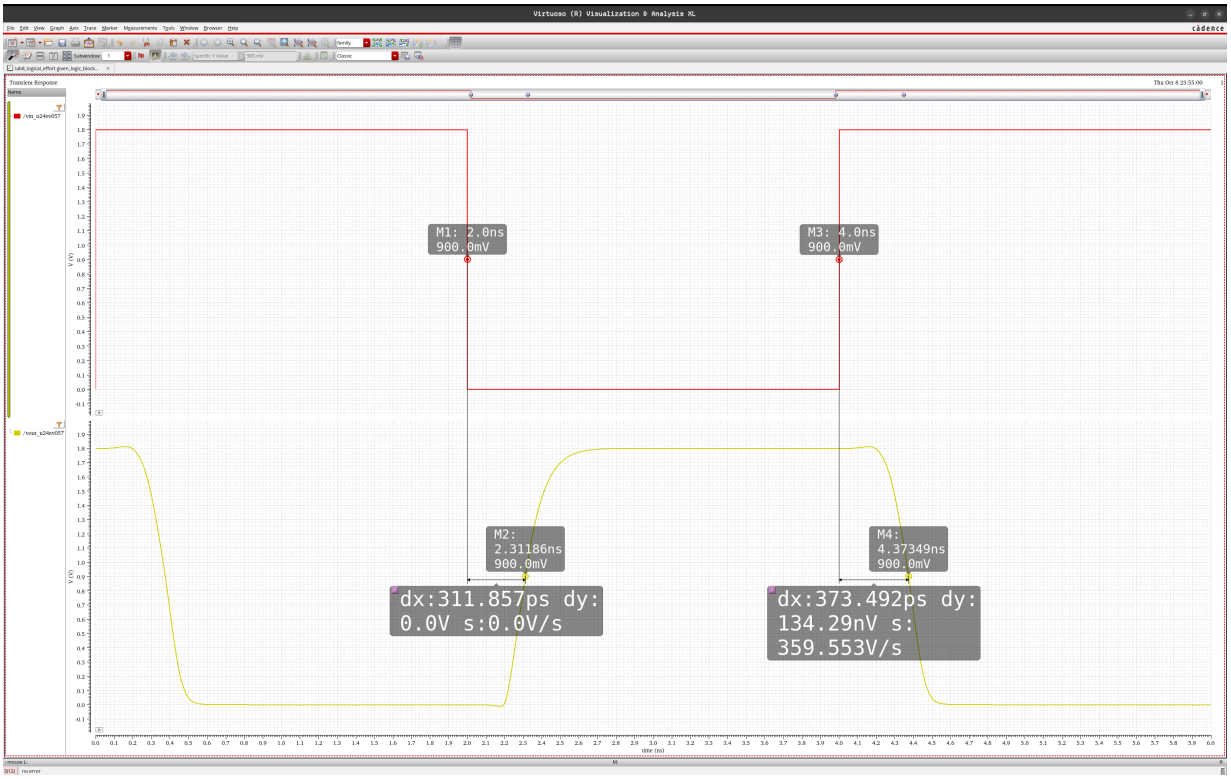


Figure 6.8: Logic block transient response waveform

$t_{pLH} = 311.857\text{ps}, \quad t_{pHL} = 373.492\text{ps}, \quad t_p = 342.6745\text{ps}$

7. CONCLUSION

1. Optimal gate sizes $x = 10$ (NAND3) and $y = 15$ (NOR2) were determined using logical effort for an optimal stage effort of $\hat{f} = 5$.

2. Simulated propagation delay (342.67ps) agreed closely with the theoretical delay (321.71ps) with $+6.52\%$ error.